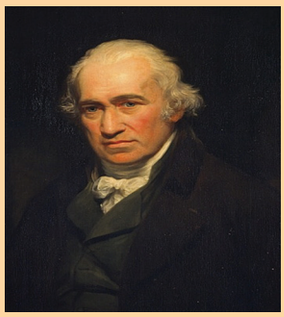
Horsepower



James Watt

The term was adopted in the late 18th century by Scottish engineer James Watt to compare the output of steam engines with the power of draft horses.

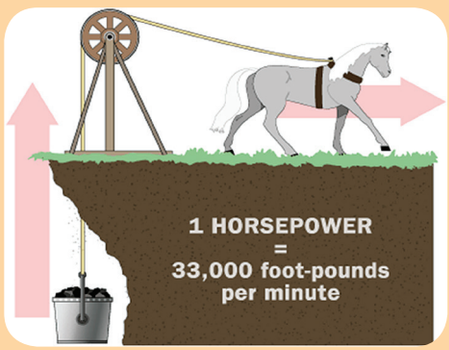
It was later expanded to include the output power of other types of piston engines, as well as turbines, electric motors and other machinery. When the steam engine began to do the work of horses in the mines during the early 1800s, the mine owners began to ask how many horses an engine would replace.

Watt measured the capability of a big horse to pull a load and found it could pull a weight of 150-pounds while walking at 2.5 miles per hour.

This works out to 33,000 foot-pounds per minute or 550 foot-pounds per second.

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The definition of the unit varied among geographical regions. Most countries now use the SI unit watt for measurement of power.



In 1900, a steering wheel is designed to replace the steering tiller and the vehicle was used for road transportation. In 1906, the first automobile was produced and sold. In 1908, henry ford manufactured 20000 cars. A lot of research and development had been made from 1910. In 1920, spark plug engine, water cooled engines are introduced.

Later, many vehicle manufacturers start begins to manufacture and sell their vehicle on the market. Many models based on the utility and usage has been introduced. Some of them include two-wheeler, three wheelers, passenger cars, luxury cars, buses, trucks etc.

4.2 ENGINE

The engine is the power plant of an automobile. A device which is used to convert one form of energy into mechanical energy is called an Engine. The heat energy produced by burning of the fuel is converted into mechanical power, then it is called a Heat Engine.

Heat engines are classified into two types.

1. External Combustion Engine
2. Internal Combustion Engine

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